CS52 · AI & ML · CIE-1 · PREDICTED QUESTIONS

Unit III: Classification & Regression

CR's 3 questions first: Classification vs regression · Naive Bayes · Linear regression with example — then 3 application problems

SURE SHOT named in the insider list / CR list or repeated from last year — learn word-perfect

VERY LIKELY came last year or is the natural variant

IMPORTANT TOPIC fits the "scenario" hint — study the idea

III Unit III — Classification & Regression (CR's list first)

YOUR CR'S 3 QUESTIONS (TEACHER SAID TO STUDY THESE FOR CIE-1)

1. Difference between classification and regression. 2. Naive Bayes classifier. 3. Explanation of linear regression with example.

Q1. Differentiate between classification and regression with examples.

SURE SHOT — CR

Both are **supervised learning** (trained on labelled data). **Classification** predicts a discrete category / class label ("Which class?"), e.g. spam / not spam. **Regression** predicts a continuous numeric value ("How much?"), e.g. house price.

Basis	Classification	Regression
Output	Discrete label (+ class probability)	Continuous real number
Question	Which category?	How much / how many?
Example (students)	Pass / Fail	Marks scored
Example (health)	Disease present / absent	Blood-glucose level
Example (real estate)	Budget / mid / premium	Selling price
Algorithms	Logistic regression, Naive Bayes, SVM, decision tree, KNN	Linear, polynomial regression, SVR, regression tree
Loss	Cross-entropy, hinge	Squared / absolute error
Metrics	Accuracy, precision, recall, F1, confusion matrix, ROC-AUC	MAE, MSE, RMSE, R^2
Picture	Decision boundary separating classes	Best-fit line / curve through points
Threshold	Converts probability → class (e.g. 0.5)	Not needed

BONUS LINE EXAMINERS LIKE

Logistic regression is a **classification** algorithm despite its name — it outputs a probability (0–1) that a threshold turns into a class.

REMEMBER IT

"Can I average two answers?" Average of ₹40 L and ₹60 L = ₹50 L ✔ → regression. Average of "cat" and "dog" ✘ → classification. **Units (₹, kg, °C, marks) → regression.**

Q2. Explain the Naive Bayes classifier with an example. SURE SHOT — CR

DEFINITION (1 MARK)

A supervised **probabilistic classifier** based on **Bayes' theorem** that predicts the class with the highest posterior probability. "**Naive**" because it assumes the features are conditionally independent given the class.

$$P(C \mid X) = \frac{P(X \mid C)\,P(C)}{P(X)} \quad \Rightarrow \quad \hat{C} = \operatorname{argmax}_C P(C) \prod_i P(x_i \mid C) \quad (P(X) \text{ is common} \rightarrow \text{dropped})$$

Email	"offer"	"meeting"	Class
E1–E3	Yes	No	Spam
E4	Yes	Yes	Spam
E5	No	No	Spam
E6–E8	No	Yes	Not
E9	Yes	Yes	Not
E10	No	No	Not

New email: offer = Yes, meeting = No
Priors: P(Spam) = P(Not) = 5/10 = 0.5
Spam: P(offer=Y) = 4/5, P(meeting=N) = 4/5
Not: P(offer=Y) = 1/5, P(meeting=N) = 1/5
Score(Spam) = 0.5 × 0.8 × 0.8 = 0.32
Score(Not) = 0.5 × 0.2 × 0.2 = 0.02
⇒ **Spam**, P = 0.32/0.34 = 94.12%

Zero-frequency + Laplace smoothing (1 mark): if a feature never occurs with a class its probability is 0 and wipes out the product → use (count + 1)/(class count + k). Here: 5/7 and 2/7 → Spam 86.21%. **Types:** Gaussian, Multinomial, Bernoulli, Categorical. **Pros:** simple, fast, good for text, works with little data. **Cons:** independence assumption, zero frequency, overconfident probabilities.

REMEMBER IT

"Prior × Likes × Likes … pick the biggest." 5 steps: **P-L-M-C-N** = Priors → Likelihood table → Multiply → Compare → Normalize. "Naive" = treats every word as a stranger to the others.

Q3. Explain linear regression with an example. SURE SHOT — CR

Linear regression is a supervised method that models a continuous dependent variable y as a straight-line function of an independent variable x: $\hat{y} = b_0 + b_1 x$ (b_0 = intercept, b_1 = slope). The best line minimises the sum of squared errors (least squares):

$$b_1 = \frac{\Sigma xy - (\Sigma x)(\Sigma y)/n}{\Sigma x^2 - (\Sigma x)^2/n} \qquad b_0 = \bar{y} - b_1 \bar{x}$$

Hours x	Marks y	x^2	xy
1	40	1	40
2	50	4	100
3	65	9	195
4	70	16	280
5	85	25	425
15	**310**	**55**	**1040**

$\bar{x} = 3, \bar{y} = 62$
$S_{xy} = 1040 - 15 \times 310/5 = 110$
$S_{xx} = 55 - 225/5 = 10$
$b_1 = 110/10 = 11, \quad b_0 = 62 - 33 = 29$
$\hat{y} = 29 + 11x$ → +11 marks per study hour
Predict 6 hours: $29 + 66 = 95$
SSE = 20, SST = 1230 → $R^2 = 1 - 20/1230 = 0.98$ (excellent fit)

Uses: sales forecasting, price prediction, risk assessment. **Assumptions:** linear relationship, independent errors, constant variance. **Limitations:** outliers, non-linear data, unreliable extrapolation. Draw the scatter plot with the line (1 mark).

REMEMBER IT

"**Slope = how much y moves per x; intercept = where it starts.**" Table columns to draw: **x | y | x^2 | xy** ("X-Y-Square-Product"). The formula's numerator uses xy, the denominator uses x^2.

Q4. Scenario: classify each problem as classification or regression and name a suitable algorithm. IMPORTANT TOPIC

Scenario	Type	Algorithm
Predict tomorrow's maximum temperature	Regression	Linear regression
Detect whether a transaction is fraud	Classification (binary)	Logistic regression / Naive Bayes
Predict a student's CGPA from attendance and hours	Regression	Multiple linear regression
Tumour benign or malignant	Classification	SVM / logistic regression
Estimate a car's resale price	Regression	Linear / polynomial regression
Classify news into sports / politics / tech	Classification (multiclass)	Naive Bayes (multinomial)
Predict number of app downloads next month	Regression	Linear regression
Will a customer churn (yes/no)?	Classification	Logistic regression / decision tree

TIP

Justify each in one phrase: "output is a number" or "output is a category". That phrase is where the mark is.

Q5. Naive Bayes numerical on a new dataset (PlayTennis / stolen car type). IMPORTANT TOPIC

If NB comes as a problem instead of theory, it will be this format.

Stolen car: 10 cars, 5 stolen. Classify (Red, SUV, Domestic).
Yes: Red 3/5, SUV 1/5, Domestic 2/5. No: Red 2/5, SUV 3/5, Domestic 3/5.

$P(\text{Yes}) \cdot \ldots = 0.5 \times 3/5 \times 1/5 \times 2/5 = 0.024$
$P(\text{No}) \cdot \ldots = 0.5 \times 2/5 \times 3/5 \times 3/5 = 0.072$
⇒ **Not stolen**

PlayTennis (14 days, 9 yes / 5 no). Classify (Sunny, Cool, High, Strong).

Yes: $9/14 \times 2/9 \times 3/9 \times 3/9 \times 3/9 = 0.0053$
No: $5/14 \times 3/5 \times 1/5 \times 4/5 \times 3/5 = 0.0206$
⇒ **No** ($P = 0.0206/0.0259 = 0.795$)

EXAM LAYOUT (GETS FULL MARKS)

(1) Prior table · (2) likelihood table only for the values in the new instance · (3) multiply for each class · (4) compare and box the class · (5) normalize. Write fractions first, decimals at the end — fewer rounding errors.

Q6. A shop's ad spend (x, ₹ thousands) and sales (y, ₹ lakhs) are given. Fit a regression line, interpret it and predict. IMPORTANT TOPIC

x	y	x − x̄	y − ȳ	(x−x̄)(y−ȳ)	(x−x̄)²
1	3	−2	−0.6	1.2	4
2	4	−1	0.4	−0.4	1
3	2	0	−1.6	0	0
4	4	1	0.4	0.4	1
5	5	2	1.4	2.8	4
x̄ = 3, ȳ = 3.6				**4**	**10**

$b_1 = 4/10 = 0.4$; $b_0 = 3.6 - 1.2 = 2.4$

$\hat{y} = 2.4 + 0.4x$

Interpretation: every extra ₹1000 of ads adds ₹0.4 lakh sales; with zero ads, sales ≈ ₹2.4 lakh.

Predict x = 6: 2.4 + 2.4 = ₹4.8 lakh (extrapolation)

$R^2 = \Sigma(\hat{y}-\bar{y})^2/\Sigma(y-\bar{y})^2 = 1.6/5.2 \approx 0.31$ → weak fit: ad spend explains only ~31% of sales variation.

REMEMBER R^2

R^2 = "**how much of the up-and-down in y the line explains**". Near 1 = line hugs the points; near 0 = points scattered.

FINAL 10-MINUTE CHECKLIST BEFORE ENTERING THE HALL

- ☐ PEAS of a self-driving car + environment type in one line
- ☐ A*: f = g + h; the 8-puzzle answer is 5 moves (U, U, L, D, R)
- ☐ Modus Ponens valid; affirming the consequent invalid; validity ≠ truth
- ☐ Curiosity chain: Dog → Lover → No-kill → Cat is Animal → Not Jack → Curiosity
- ☐ Teacher / Explorer / Gamer → S / U / R
- ☐ Bayes: G-B-T-A; the disease example gives ≈ 8.8%
- ☐ Classification = category, regression = number (+ the 10-row table)
- ☐ Naive Bayes: P-L-M-C-N; email → Spam 94.12%
- ☐ Linear regression: x | y | x² | xy → ŷ = 29 + 11x